

Code Layout, Readability and Reusability

Project Title

AI Learning Assistant

1. Introduction

A well-organized code structure improves software quality, readability, maintenance, and future scalability. The AI Learning Assistant follows a modular architecture where each module performs a specific task independently.

2. Project Folder Structure

AI-Learning-Assistant/

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|— app.py

|— emotion_detector.py

|— gemini_helper.py

|— logger.py

|— dashboard.py

|— requirements.txt

|— .env

|

└─ data/
 └─ emotion_logs.csv

3. Module Description

File	Purpose
app.py	Main Streamlit application
emotion_detector.py	Detects user emotions
gemini_helper.py	Connects with Gemini AI
logger.py	Stores interaction history
dashboard.py	Displays analytics
requirements.txt	Project dependencies

4. Readability

The project follows coding best practices:

- Meaningful variable names
- Proper comments
- Modular functions
- Simple code structure
- Consistent formatting

5. Code Reusability

Reusable modules include:

- Emotion Detection Module
- Gemini API Module
- Logger Module
- Dashboard Module

These modules can be reused in future AI-based applications.

6. Advantages

- Easy maintenance
 - Better debugging
 - Reusable components
 - Scalable architecture
 - Improved collaboration
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7. Conclusion

The modular code layout improves readability, maintainability, and software quality while supporting future enhancements.